Object Detection

Finding and Classifying Objects in Images

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1. What Object Detection Is

Object detection is the computer vision task of finding **where** objects are in an image (localization, usually a bounding box) and **what** they are (classification), for potentially multiple objects in a single image. It sits between image classification (single label, whole image) and image segmentation (pixel-level labeling).

2. Output Format

For each detected object, a model typically outputs:

- A bounding box, usually (x_min, y_min, x_max, y_max) or (center_x, center_y, width, height).
- A class label (e.g., 'dog', 'car').
- A confidence score reflecting how certain the model is.

3. Two-Stage Detectors

Two-stage detectors first generate candidate regions likely to contain an object, then classify and refine each one. The R-CNN family is the classic example:

- **R-CNN**: uses an external algorithm (Selective Search) to propose ~2000 regions, runs a CNN on each crop independently - accurate but very slow.
- **Fast R-CNN**: runs the CNN once on the whole image and pools features per region (RoI Pooling), sharing computation across proposals.
- **Faster R-CNN**: replaces the external proposal algorithm with a learned Region Proposal Network (RPN), making the whole pipeline trainable end-to-end.

Note: See the companion 'Region Proposal' document for details on how the RPN itself works.

4. One-Stage Detectors

One-stage detectors skip the separate proposal step and predict boxes and classes directly in a single pass, trading some accuracy for large speed gains.

- **YOLO (You Only Look Once)**: divides the image into a grid; each cell predicts boxes, objectness, and class probabilities in one forward pass.
- **SSD (Single Shot Detector)**: predicts from multiple feature-map scales simultaneously to handle objects of different sizes.

- **RetinaNet**: introduces Focal Loss to address the extreme foreground/background class imbalance that hurts one-stage detectors.

5. Transformer-Based Detectors

DETR (Detection Transformer) reframes detection as a direct set-prediction problem: a CNN backbone extracts features, a transformer encoder-decoder attends over them, and a fixed set of learned 'object queries' each predict one object (or 'no object'), removing the need for hand-designed anchors or non-max suppression.

6. Key Building Blocks

- **Anchor boxes**: predefined boxes of various sizes/aspect ratios tiled across the image, used as reference points that the network learns to adjust (used in Faster R-CNN, SSD, YOLOv2-v4).
- **Non-Max Suppression (NMS)**: post-processing step that removes duplicate overlapping boxes for the same object, keeping only the highest-confidence one.
- **Feature Pyramid Network (FPN)**: combines features from multiple resolutions so the model detects both small and large objects well.

7. Evaluation: IoU and mAP

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IoU = area(box_predicted AND box_ground_truth) / area(box_predicted OR box_ground_truth)
```

A prediction is counted as correct if its IoU with a ground-truth box exceeds a threshold (commonly 0.5). **mean Average Precision (mAP)** then summarizes precision/recall across all classes and confidence thresholds into a single benchmark number.

8. Common Applications

- Autonomous driving (pedestrians, vehicles, signs).
- Retail and inventory (shelf and product detection).
- Medical imaging (tumor or anomaly localization).
- Surveillance and video analytics.